

Some Possible Numerical Problems Based on the Syllabus of Data Science and Analytics

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This document describes different types of problems that could be asked in the theory of exam for the course Data Science and Analytics (BE CMP/IT, PU).

General note for numerical problems: The primary objective of numerical problems should be to evaluate how well students have grasped the underlying concepts presented in the lectures. Accordingly, the questions should emphasize conceptual understanding rather than involve extensive calculations—since repetitive computations are better suited to computers than to humans. As a result, wherever appropriate, problems should be designed using a minimal number of data points to keep the focus on reasoning and interpretation.

Unit 1

This unit can cover numerical problems related to the discretization of continuous variables. To make problems tractable, the number of data points provided should not exceed twenty.

Below are some examples of numerical problems for this unit.

1. A numeric variable has the following values.
11.49, 9.59, 12.94, 16.57, 8.30, 8.30, 16.74, 13.30, 6.31, 10.83, 10.47, 10.64, 13.24, 7.09, 7.28, 13.44, 9.99, 8.09, 9.07, 6.30
Discretize the values in five levels based on the frequencies.
2. A numeric variable has the following values.
46.93, 138.07, 64.77, 51.46, 15.38, 15.37, 11.32, 223.21, 120.23, 19.85, 19.80, 12.39, 55.79, 86.31, 3.99, 101.51, 30.71, 94.28, 102.39, 18.04
Standardize the values.

Unit 2

Numerical problems in this unit may require students to visually interpret different empirical distributions through graphical analysis. Additionally, questions can be designed to assess students' understanding of key statistical concepts, including various types of correlation, p-values, and chi-squared statistics.

Below are some examples.

1. Based on a survey, the variable x takes the following values.
0.76, 1.53, 0.12, 2.44, 2.66, 3.84, 1.03, 6.31, 2.03, 4.61, 9.53, 10.39, 3.69, 6.63, 6.4, 5.01, 2.99, 1.18, 0.35, 0.44, 1.99, 7.04, 3.05, 7.89, 1.44, 1.11, 7.53, 10.4, 1.72, 4.96
Do you think x follows a normal distribution or an exponential-type distribution? Support your answer with appropriate plots.
2. The following table presents a hypothetical cross-tabulation between 'food preferences' and 'gender'.

	Female	Male	Other
Likes buff momo	8	75	1
Likes chicken momo	63	3	0
Likes veg momo	23	19	3

Do you think food preferences are correlated with gender? Justify using chi-squared test with significance level 0.05. The critical value for the Chi-Squared statistic with 4 degrees of freedom at a significance level of 0.05 is 9.4877.

3. Below are the values of two covariates x_1 and x_2 .

x_1	x_2
0.50	0.25
-0.14	-0.43
0.65	0.94
1.52	1.64
-0.23	-0.37
-0.23	-0.18
1.58	1.36
0.77	0.87
-0.47	-0.60
-0.47	-0.41

Calculate the Karl Pearson correlation coefficient between x_1 and x_2 .

4. Suppose you surveyed 5 male students and found their average daily spending on refreshments is Rs. 730, with a sample standard deviation of Rs. 200. A previous study shows that female students spend Rs. 810 per day on average. Using a t-test, determine if male students' daily spending is significantly different from that of female students. (Use $|t| > 1.8$ as the threshold for significance.)

Unit 3

For this unit, numerical problems may cover measures of fit, multicollinearity (VIFs), and the Nadaraya-Watson estimator. While numerical problems related to PCA may not be practical for exams, questions focused on the interpretation of principal components can still be included.

Below are some examples.

1. The following table shows the values of variables x_1 , x_2 , and y . The fourth column shows the estimated value of y obtained by regressing y on x_1 and x_2 .

x_1	x_2	y	Fitted $\hat{y}$
1.0	2.0	8.1	8.0
2.0	1.5	8.4	8.5
1.5	2.5	11.3	11.0
2.5	2.0	11.7	11.5
3.0	1.0	9.2	9.0
1.2	1.8	8.3	8.4
2.2	2.2	12.2	12.0
1.8	2.0	10.0	10.2
2.8	1.5	10.8	10.6
2.0	2.5	12.9	12.5

Calculate the values of R-squared and Adjusted R-squared for the regression model.

2. Suppose that you are estimating a linear model

$$y = a_0 + a_1x_1 + a_2x_2 + a_3x_3$$

Now, you decided to calculate the variance inflation factor for the variable x_3 . Following the usual procedure, you regressed x_3 on x_1 and x_2 with the results summarized in the following table.

x_1	x_2	x_3	Fitted $\hat{x}_3$
1.0	2.0	6.2	6.5
2.0	1.5	6.4	6.75
1.5	2.5	8.9	8.75
2.5	2.0	8.6	8.75
3.0	1.0	7.0	7.0
1.2	1.8	6.0	6.3
2.2	2.2	9.5	9.35
1.8	2.0	8.0	8.1
2.8	1.5	8.2	8.25
2.0	2.5	9.9	9.75

What is the variance inflation factor for x_3 ?

3. You are given the following data.

x	1	2	3	4
y	-2.3	0.5	9.3	20.8

Assuming that there is an unknown functional relationship $y = f(x)$, estimate the value of y at $x = 2.9$ using the Nadaraya-Watson regression estimator. Take bandwidth $h = 0.5$.

Unit 4

Numerical problems in this unit may focus on the interpretation of logistic regression models and the iterations of the k-means algorithm. As noted earlier in the general guidelines, it is impractical to ask students to compute more than one iteration of the k-means algorithm. Additionally, in the context of clustering, numerical questions involving similarity scores based on common distance metrics may also be included.

Below are some examples.

1. A researcher relates the probability of passing an exam and hours studied per day by a student by estimating the following logistic regression model.

$$\log\left(\frac{p}{1-p}\right) = -4 + 1.2x$$

where p is the probability of passing an exam and x is the hours studied.

- What is the predicted probability of passing if a student studies for 3 hours?
- Interpret the coefficient of x , i.e. hours studied.

2. There are objects with attributes x_1 and x_2 as given in the following table.

Item	x1	x2
A	5	3
B	-1	1
C	1	-2
D	-3	-2

You are using the k-means clustering algorithm with 2 clusters ($k = 2$) to group four items: A, B, C, and D. At the start (initial assignment):

- Items A and B are placed in Cluster 1
- Items C and D are placed in Cluster 2

After running one iteration of the k-means algorithm, which cluster will each item belong to?

Unit 5

Since the concepts covered in this unit are relatively advanced, it may not be practical to include numerical problems for every topic. However, questions related to the fundamentals of autocorrelation and stationarity can still be asked. An example is provided below.

1. You are given two series, A and B , with following values for time $t = 1, 2, \dots, 6$.

Time, t	1	2	3	4	5	6
A	5	7	9	11	13	15
B	3	7	5	8	6	7

Which of these two series is more likely to be stationary? Justify your answer.